Statistics & Sampling

ANSWER KEY

Sampling methods

1 A school wants to survey students about the cafeteria. Explain why surveying only students in the chess club would produce biased results, and suggest a better sampling method.

Surveying only chess club members would not represent the full student body — their opinions might differ systematically from other students (e.g. different schedules, different cafeteria usage patterns), introducing bias. A better method: use random sampling, such as randomly selecting students from the full enrollment list across all grades and interests, so every student has an equal chance of being surveyed.

Comparing distributions using shape, center, and spread

2 Two data sets have the same median but Data Set A has a much larger range than Data Set B. Explain what this tells you about the variability of each data set.

Since Data Set A has a larger range despite having the same median, its data values are more spread out (more variable) than Data Set B's — the values in Data Set A extend further from the center, even though both sets are centered around the same typical value.

Linear regression and correlation

3 A scatterplot of hours studied vs. exam score has a line of best fit $y = 6.5x + 52$. Interpret the slope and y-intercept in context.

The slope, 6.5, means each additional hour of studying is associated with an increase of about 6.5 points on the exam score. The y-intercept, 52, represents the predicted exam score for a student who studies 0 hours.

4 A data set has a correlation coefficient of $r = -0.85$. Describe the strength and direction of the relationship.

Since r is negative, the relationship is a negative (inverse) association — as one variable increases, the other tends to decrease. Since $|r| = 0.85$ is close to 1, the relationship is strong.

Using a line of best fit to make predictions

5 A line of best fit for predicting weight (in kg) from height (in cm) is $W = 0.9H - 70$. Predict the weight of a person who is 175 cm tall, and explain a limitation of using this model for a person who is 250 cm tall.

$W = 0.9(175) - 70 = 157.5 - 70 = 87.5$ kg. Using the model for a height of 250 cm would be an extrapolation far outside the range of the original data used to build the model, so the prediction may be unreliable — the linear relationship observed for typical human heights may not hold at such an extreme, unrealistic value.

Interpreting the standard deviation

6 Data Set A has standard deviation 2, and Data Set B has standard deviation 8, both with the same mean. Explain what this indicates about the two data sets, and give a real-world example of each.

A smaller standard deviation (Data Set A) means the data values are clustered closely around the mean, while a larger standard deviation (Data Set B) means the values are more spread out. Example: test scores on an easy, uniform quiz might have a small standard deviation (most students score similarly), while test scores on a challenging exam with mixed preparation levels might have a large standard deviation (scores vary widely).

Margin of error and confidence in survey results

7 A survey estimates that 42% of a population favors a new policy, with a margin of error of 3%. A second survey with a larger sample size estimates the same statistic with a margin of error of 1.5%. Explain why the second survey's margin of error is smaller, and what this means for the precision of the estimate.

A larger sample size generally produces a smaller margin of error, because a bigger sample gives a more precise (less variable) estimate of the true population value. The second survey's narrower margin of error (1.5% vs. 3%) means its estimate is more precise — the true population percentage is likely confined to a tighter range around the reported statistic.

Evaluating experimental design

8 A researcher wants to test whether a new fertilizer increases crop yield. Explain why using a control group (plants without the fertilizer) is essential to this experiment's design.

A control group provides a baseline for comparison — without it, the researcher cannot determine whether any observed increase in crop yield is actually caused by the fertilizer, or whether it would have happened anyway due to other factors (weather, soil quality, time). Comparing the treatment group (fertilizer) to the control group (no fertilizer) under otherwise identical conditions isolates the fertilizer's specific effect.

Calculating and comparing means from grouped data

9 A class of 25 students has a mean test score of 78. A second class of 15 students has a mean score of 88. Find the combined mean score of all 40 students.

Total points from class 1: $25 \times 78 = 1950$. Total points from class 2: $15 \times 88 = 1320$. Combined total: $1950 + 1320 = 3270$. Combined mean: $\frac{3270}{40} = 81.75$.

Recognizing bias in survey questions and methods

10 A survey asks, "Don't you agree that the excellent new park should be funded?" Explain why this question is biased, and rewrite it in a neutral form.

The question is biased because it uses leading language ("Don't you agree" and "excellent") that pressures respondents toward a particular answer, rather than allowing them to form an independent opinion. A neutral version: "Do you support funding the new park? Yes or No."

Comparing two data sets using boxplot-style summaries

11 Data Set A has minimum 10, Q1 = 20, median 30, Q3 = 40, maximum 50. Data Set B has minimum 15, Q1 = 28, median 30, Q3 = 32, maximum 45. Both sets have the same median. Compare their spread using the interquartile range (IQR).

IQR of Set A: $Q3 - Q1 = 40 - 20 = 20$. IQR of Set B: $Q3 - Q1 = 32 - 28 = 4$. Since Set A has a much larger IQR, its middle 50% of data is far more spread out than Set B's, even though both sets share the same median.

Extrapolation vs. interpolation

12 A line of best fit is built from data on shoe size (ranging from 6 to 12) and height. Explain the difference between using this model to predict height for a shoe size of 9 versus a shoe size of 20, referencing interpolation and extrapolation.

Predicting height for a shoe size of 9 is interpolation, since 9 falls within the original data range (6 to 12) — this is generally reliable. Predicting for a shoe size of 20 is extrapolation, since it falls far outside the observed data range — this is much less reliable, since the linear relationship observed within the data range may not hold true for such an extreme, unobserved value.

Relative frequency and probability from a frequency table

13 A frequency table of exam grades shows: A: 8 students, B: 15 students, C: 12 students, D: 5 students. Find the relative frequency (as a percent) of students who earned a B or higher.

Total students: $8 + 15 + 12 + 5 = 40$. Students with B or higher: $8 + 15 = 23$. Relative frequency: $\frac{23}{40} = 0.575$, or 57.5%.

Random assignment vs. random sampling

14 Explain the difference between random sampling and random assignment, and state which one allows a study to make causal claims.

Random sampling refers to how participants are SELECTED from a population, which allows results to be generalized to that population. Random assignment refers to how selected participants are ASSIGNED to treatment vs. control groups, which allows a study to make causal claims — because random assignment balances out other confounding factors between groups, any resulting difference can be attributed to the treatment itself rather than pre-existing group differences.

A multi-part regression and residual analysis problem

15 This question has three parts. A line of best fit for predicting sales revenue (in thousands of dollars) from advertising spend (in thousands of dollars) is $R = 4.2A + 18$. (a) Predict the revenue for an advertising spend of 10 thousand dollars. (b) If the actual observed revenue at $A = 10$ was 55 thousand dollars, find the residual (actual minus predicted). (c) Explain what a positive residual indicates about this particular data point relative to the line of best fit.

(a) $R = 4.2(10) + 18 = 42 + 18 = 60$ thousand dollars predicted. (b) Residual $= \text{actual} - \text{predicted} = 55 - 60 = -5$ thousand dollars. (c) Since the residual here is actually negative (-5), this indicates the actual revenue was lower than the model predicted — this specific data point lies below the line of best fit. A positive residual, in general, would indicate the actual value exceeds the model's prediction, meaning the point lies above the line.